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A1 Data, Splits and Sample Sizes

Same chronological boundaries for every candle length; test scored once

Candle	Rows (all)	Train	Rows fitted	Validation	Test	Test dojis	Test bullish share
1 min	5,884,299	3,167,361	2,000,000	1,346,246	1,370,692	58,616	50.4%
5 min	1,312,758	770,236	653,403	269,182	273,340	4,933	50.8%
15 min	442,614	264,084	240,005	88,341	90,189	909	51.3%
1 hour	108,026	64,877	62,096	21,642	21,507	104	51.8%
1 day	3,457	1,991	1,983	822	642	0	54.2%

- Train 2006-01-03 → 2017-08-24, validation → 2021-07-14, test 2021-07-15 → 2025-05-30; labels crossing a boundary are purged. Test seen in earlier phases: exploratory.
- Data: TradeStation continuous NQ 1-minute bars, additively back-adjusted (point moves equal the traded contract's).
- NQ = $20 × Nasdaq-100 index, tick 0.25 = $5; notional ≈ $210k–$445k per contract in 2021–2025. P&L = points moved × $20.
- Candles stay inside one CME session (daily = 17:00–16:00 CT). Bad prints (|1-min return| > 2%) quarantine the next 50 candles.
- Daily: early-close sessions are not targets; the quarantine after 3 Feb 2025 ends the daily test on 2025-01-30.

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A2 Features, Hyper-parameters and Selection Rules

All choices made on training and validation data; the test set never informed them

Features (all known at the close of candle t)

Group	Features
Original (8)	log returns lag 1, 2, 3, 5; momentum 5, 10; volatility 5, 10
Candle shape	body / close, high-low range, upper and lower wick
Volume, volatility	log volume vs 20-bar mean; 20-bar realised volatility
Indicators	RSI(14) centred; gap to 20 and 50-bar moving averages
Time	time of day within the session (sine, cosine)

Configurations tested per candle length

Family	Original	Variants (new features)
Logistic	C = 1, 8 features	C = 1 and C = 0.01
Random forest	200 trees, depth 6	200 trees, depth 10, leaf 200
XGBoost	200 trees, depth 3, η 0.05	depth 3 or 5, η 0.05, early stop on validation
MLP	64-32-16, ReLU, L2 1e-3	64-32, 128-64-32, 256-128-64-32; ±dropout 0.2 + batch norm; lr 3e-4 / batch 1024; AdamW, early stop

- Optimised model per family = highest validation balanced accuracy (it may be the original).
- Confidence threshold per model and direction = highest validation net P&L on {0, 0.01, 0.02, 0.03, 0.05, 0.075, 0.10}, counting only executable entries, at least 200 trades; if none qualifies the filtered strategy is not evaluable (no fallback to 0).
- Overall strategy = highest validation net P&L; deployable only if test net > 0 at $5 and $12.50, daily-P&L 95% interval > 0 and ≥ 3 of 4 profitable walk-forward years (not-evaluable years do not count).
- Scaling fitted on training rows only; network training shuffles only inside the training set; seeds fixed (42). Daily networks use batches of 128 (≈2,000 training rows).
- Profit factor = sum of winning trades ÷ |sum of losing trades|, after commissions. Gross / trade = average P&L per trade before the $5 commission.

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A3 Classification Metrics and Confusion Matrix

Test period, 5 min candles: originals and optimised versions

Model	Accuracy	Non-doji	Balanced	Precision (up)	Recall (up)	F1 (up)	Bullish acc.	Bearish acc.	vs majority, pp [95% CI]
Always long	49.90%	50.81%	50.00%	50.81%	100.00%	67.39%	100.00%	0.00%	0.00 [0.00, 0.00]
Reversal rule	50.30%	51.23%	51.22%	52.04%	51.27%	51.65%	51.27%	51.18%	0.40 [0.12, 0.68]
Logistic (original)	50.42%	51.35%	51.16%	51.76%	62.85%	56.76%	62.85%	39.48%	0.53 [0.25, 0.80]
Logistic C=1 + features	50.43%	51.36%	51.28%	51.99%	55.90%	53.87%	55.90%	46.67%	0.53 [0.27, 0.79]
XGBoost (original)	50.50%	51.43%	51.33%	52.00%	57.43%	54.58%	57.43%	45.23%	0.60 [0.32, 0.91]
XGBoost d3, early stop	50.46%	51.39%	51.32%	52.03%	55.73%	53.81%	55.73%	46.91%	0.57 [0.27, 0.86]
Random forest (original)	50.49%	51.42%	51.32%	52.00%	57.33%	54.53%	57.33%	45.32%	0.60 [0.30, 0.88]
Random forest d10	50.47%	51.40%	51.34%	52.07%	54.93%	53.46%	54.93%	47.75%	0.57 [0.28, 0.88]
MLP 64-32-16 (original)	50.33%	51.26%	51.12%	51.78%	59.48%	55.36%	59.48%	42.77%	0.44 [0.16, 0.71]
MLP 128-64-32	50.52%	51.45%	51.42%	52.18%	53.16%	52.67%	53.16%	49.68%	0.62 [0.32, 0.92]

Confusion matrix, MLP 128-64-32 (non-doji candles)

	Predicted up	Predicted down
Actual up	72,499	63,891
Actual down	66,431	65,586

n = 273,340 test candles (136,390 bullish, 132,017 bearish, 4,933 doji). Accuracy ± 0.19 pp (binomial 95%); the paired comparison with the majority rule uses 20-session block bootstrap.

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A4 Net P&L by Candle Length, Model and Direction

Test period, $5 per round trip, thousand USD: unfiltered (left) and with the validation-chosen filter (right)

Candle	Model (optimised)	L&S	Long	Short	L&S filtered	Long filtered	Short filtered
1 min	Logistic C=1 + features	-7715.1	-4023.4	-3691.7	-70.6	-45.5	-25.1
	Random forest d10	-7366.8	-4066.8	-3300.0	-3.3	-3.2	-65.2
	XGBoost d5, early stop	-7393.0	-4122.6	-3270.3	-14.0	-11.5	-2.5
	MLP 256-128-64-32	-7088.5	-3909.7	-3178.8	30.7	2.2	28.5
5 min	Logistic C=1 + features	-1158.5	-630.8	-527.7	36.1	-16.1	44.0
	Random forest d10	-1045.1	-560.1	-485.0	-5.5	10.1	-15.6
	XGBoost d3, early stop	-1135.4	-616.3	-519.1	29.9	14.6	15.3
	MLP 128-64-32	-1138.5	-581.6	-556.9	-26.1	-3.5	-9.8
15 min	Logistic C=0.01 + features	-557.1	-317.7	-239.4	-16.6	-36.6	20.0
	Random forest d10	-335.6	-200.6	-135.0	4.0	3.8	13.7
	XGBoost d3, early stop	-408.7	-251.7	-157.0	-15.6	-18.1	44.8
	MLP 128-64-32	-515.1	-290.3	-224.9	-52.5	-60.8	8.2
1 hour	Logistic C=1 + features	-128.9	-89.2	-39.8	-189.4	-201.3	-20.0
	Random forest d10	-5.2	-31.0	25.7	-61.4	—	-0.5
	XGBoost d5, early stop	-74.9	-69.1	-5.8	-7.6	-29.1	21.5
	MLP 128-64-32	-130.0	-92.5	-37.6	-137.6	-106.9	-30.7
1 day	Logistic (original)	110.0	92.4	17.6	92.3	68.2	—
	Random forest d10	73.0	73.0	0.0	—	—	—
	XGBoost (original)	56.9	66.1	-9.1	92.2	76.7	15.6
	MLP 64-32-16 (original)	73.0	73.0	0.0	—	—	—

"—": the validation-chosen threshold was 0 (no filter) or no threshold reached 200 validation trades. Intraday filtered results are spread around zero; daily models that call "up" every day equal always long. Descriptive: one search, many comparisons.

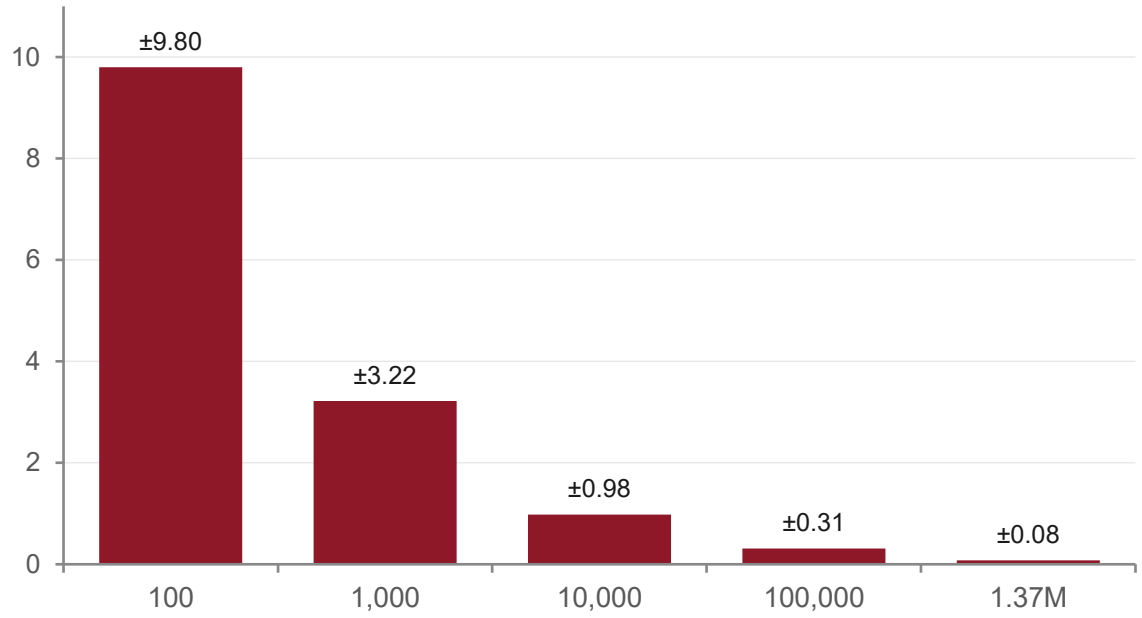
Base case: idealised fills at the next open, zero latency, 100% fills, zero slippage, $5 per round trip

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A5 Why 50% Means Random

Predicting the next candle is close to tossing a coin; only results outside luck's range count

How far luck alone moves a fair coin's accuracy (95% range, ± pp)



Range = 1.96 × √(0.25 / n). With few trades a coin can look like a 55–60% "model"; with a million it cannot leave 50 ± 0.1%.

Reading accuracy around 50%

1 **Coin toss = 50%**
if up and down are equally likely and unpredictable, any rule is right half the time

2 **Dojis lower the bar**
candles with close = open count as misses, so all-candle accuracy can sit below 50%

3 **Imbalance is not skill**
NQ rose over time, so "always up" already beats a coin without predicting anything

4 **Accuracy ≠ money**
accuracy alone does not set profit: move sizes and costs matter. With equal ±W moves, expected P&L per trade = (2p − 1)·W − $5, i.e. 0.02·W − $5 at p = 51%

Best model: 51.45% on non-doji 5 min candles: detectable but exploratory, economically too small after costs

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A6 Reproducibility and Limitations

Tables are regenerated from the run files; verify.py re-derives the test metrics from the saved predictions

Reproduce (Windows PowerShell, project folder)

Repository: github.com/Paciooo/Project_ai_ML_bigdata

```
$env:FUTURES_DATA_DIR="C:\path\to\futures-data"
```

```
1. python study.py
```

trains all models, backtests, walk-forward (about 15 min on 16 CPU cores, no GPU)

this deck: 1 min, 5 min, 15 min, 1 hour run + 1 day run, joined by tools/merge_runs.py

```
2. python tools/wf_threshold_status.py <run>   (only runs made before the threshold fix; report.py stops without it)
```

```
   python report.py results/nq_study/runs/<run> presentation/deck_data.json
```

```
3. python verify.py results/nq_study/runs/<run>
```

test accuracy, trades, P&L, win rate, profit factor, drawdown, equity, costs; it does not rebuild data, refit models or check slide text

```
4. cd presentation; npm install; node build_deck.js
```

Runs used here: 20261111T070520Z_949daa_final (commit ff445f3) + 20261111T103357Z_02bd49_daily_final (commit 0970132); seed 42, latency 0s, $5 per round trip.

Limitations

- Idealised execution: fill at the next open with zero latency and zero slippage, 100% limit fills; a live limit order may not fill.
- Bar prices, not bid/ask quotes; $12.50 / $23 are assumed cost scenarios, not measured execution.
- Drawdown is measured on session-end equity: intra-session dips are not included.
- Test-period results of non-selected configurations are descriptive: many comparisons, no correction.
- Continuous back-adjusted series; roll dates are approximate (no position spans a session).
- The 2021–2025 test period was seen in earlier phases; data after May 2025 (reserved) was not available.
- Daily Sharpe ratios use 252 sessions; percentages use an assumed $100,000 account.

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A7 1 day Walk-Forward in Detail

MLP 256-128-64-32 + dropout/BN, Long & Short, refitted each year; always long measured on exactly the same sessions

Fold (test years)	Validation sessions	Threshold	Trades (long / short)	Net, filtered	Net, unfiltered	Always long, same sessions	Gross: direction mix	Gross: residual (timing)
0: 2021-07 → 2022-07	233	0	185 (141 / 44)	+$86.7k	+$86.7k	−$29.7k	−$15.1k	+$102.7k
1: 2022-07 → 2023-06	185	**not evaluable**	—	n/e	+$34.5k	+$20.5k	—	—
2: 2023-06 → 2024-06	70	**not evaluable**	—	n/e	+$48.3k	+$65.0k	—	—
3: 2024-06 → 2025-01	236	0.03	117 (105 / 12)	+$32.0k	+$4.9k	+$17.2k	+$11.2k	+$21.5k

- Not evaluable: the validation year had fewer than 200 sessions, so no threshold could be validated. Code up to commit 35be806 silently traded those years unfiltered and reported 4 of 4 positive years, +$201.5k; the corrected rule counts only evaluable years (2 of 2, +$118.7k).
- Direction mix = (longs − shorts) × mean move × $20: what a random policy with the same numbers of longs and shorts would gross on the same sessions; residual = the rest. Published four years: gross +$204.5k, of which +$67.1k direction mix and +$137.4k residual, vs +$73.0k for always long on the same sessions.
- This does not show “profit = being long in a rising market”: in fold 0 always long lost −$29.7k while the refitted model earned +$86.7k unfiltered. Nor does it show an edge: one path, one configuration chosen on validation among 407, idealised fills, and the same model fitted once loses −$25.9k on the test period.

Base case: idealised fills at the next open, zero latency, 100% fills, zero slippage, $5 per round trip

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